About 2,600 results for “review wildfire spread prediction”

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AdriaFirePropagator and AdriaFireRisk: user friendly Web based wildfire propagation and wildfire risk prediction software

M. Bugarić D. Stipaničev Toni Jakovcevic Environmental Science, Computer Science · 9 November 2018

TLDR The HOLISTIC Project with full name Adriatic Holistic Forest Fire Protection was a project to establish user friendly and simple regional **wildfire** propagation and **wildfire** risk **prediction** simulation tools that could be easily used by fire fighters in everyday fire fighting operative work, but also as a tool for fire season preparation. Expand

2 PDF Publisher Save Cite

Coupled Dynamic Data-Driven Framework for Forest Fire Spread Prediction

C. Brun A. Cortés T. Margalef Environmental Science, Computer Science · International Conference on Dynamic Data-Driven… · 5 November 2014

TLDR A Dynamic Data-driven methodology is used that is a calibration stage previous to the forecast where complementary models, data injection and intelligent systems are working in a symbiotic way to reduce the forecast errors at real time. Expand

5 Publisher Save Cite

Predicting wildfire spread and behaviour in Mediterranean landscapes

M. Salis B. Arca +13 authors D. Spanò Environmental Science · 27 October 2016

The use of spatially explicit fire **spread** models to assess fire propagation and behaviour has several applications for fire management and research. We used the FARSITE simulator to predict the… Expand

61 PDF Publisher Save Cite

Fire Spread Modeling Using Probabilistic Cellular Automata

Rohit Ghosh Jishnu Adhikary R. Chemlal Environmental Science, Computer Science · ASCAT · 15 February 2024

TLDR The model introduces an enhanced CA-based methodology for **wildfire prediction**, emphasizing interactions between neighboring cells and incorporating major determinants of fire **spread**, including wind direction, wind speed, and vegetation density, while also accounting for spotting and probabilistic transitions between states in the model to mirror real-world fire behavior. Expand

3 [PDF] arXiv Save Cite

Spain on Fire: A novel wildfire risk assessment model based on image satellite processing and atmospheric information

Helena Liz-L'opez Javier Huertas-Tato Jorge P'erez-Aracil C. Casanova-Mateo J. Sanz-Justo David Camacho

Environmental Science, Computer Science · Knowledge-Based Systems · 8 June 2023

TLDR The WAM uses a residual-style convolutional network architecture to perform regression over atmospheric variables and the greenness index, computing necessary resources, the control and extinction time, and the expected burnt surface area to anticipate the economic and ecological impact of a **wildfire**. Expand

13 [PDF] arXiv Save Cite

Sources, diffusion and prediction in COVID-19 pandemic: lessons learned to face next health emergency

M. Coccia · Medicine, Environmental Science · AIMS Public Health · 2 March 2023

TLDR Results show the vital role of environmental factors in the **spread** of Coronavirus Disease 2019 (COVID-19), and many limitations of the epidemiologic models of **prediction** because of the complex interactions between the new viral agent SARS-CoV-2, environment and society that have generated variants and sub-variants with rapid transmission. Expand

85 [PDF] PDF Save Cite

A Review on the Role of Machine Learning in Predicting the Spread of Infectious Diseases

S. S. Mohamed Elkanzi · Computer Science, Medicine · Metaheuristic Optimization **Review** · 2024

TLDR Evidence for how AIML has revolutionized infectious disease management is provided, calls for responsible innovation and ethical deployment of AI, and encourages international collaborations to safeguard the global health sector against new and emerging diseases are called for. Expand

3 Publisher Save Cite

A Generative Model for Surrogates of Spatial-Temporal Wildfire Nowcasting

Sibo Cheng Yike Guo Rossella Arcucci · Environmental Science, Computer Science · IEEE Transactions on Emerging Topics in... · 5 August 2023

TLDR A generative model is proposed using a three-dimensional Vector-Quantized Variational Autoencoders to generate spatial-temporal sequences of unseen **wildfire** burned areas in a given ecoregion, which succeeds in generating coherent and structured fire scenarios. Expand

10 [PDF] IEEE Save Cite

Artificial Intelligence for Antimicrobial Resistance Prediction: Challenges and Opportunities towards Practical Implementation

Tabish Ali Sarfaraz Ahmed Muhammad Aslam · Medicine, Computer Science · Antibiotics · 1 March 2023

TLDR The paper concludes with the encouragement to apply AI to the AMR sector with the intention of practical diagnosis and treatment, since presently most studies are at early stages with minimal application in the practice of diagnosis and Treatment of disease. Expand

88 [PDF] PDF Save Cite

Wildfire and power grid nexus in a changing climate

Soroush Vahedi Junbo Zhao +5 authors Bin Wang · Environmental Science, Engineering · Nature Reviews Electrical Engineering · 24 March 2025

| Global **wildfire** events have caused escalating impacts all over the world in recent years, particularly in the western United States, due to extreme fire-weather, fuel accumulation, and numerous... Expand

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